

Milestone 2.4: Dimensionality Reduction

INFO 521 · Machine Learning Foundations · project milestone brief ·
From Inference to Discovery · Gate B (Unsupervised) · 2 of 2

How to read this

This is the assignment brief for one project milestone. The Satisfactory criteria below are authoritative: every one must be met, and it is all-or-nothing rather than partial credit. A Not-Yet returns with targeted feedback and one revise cycle. Do the work in the notebook distributed through Classroom 50; this PDF is the brief, not the workspace.

Course-schedule placement. Part 2 of the project: begins at the midpoint, due Week 7.5. This milestone covers M7 and is due Week 7. Gated by Gate B, with 2.3 (Checkpoint quiz 2.3).

The structure lens

Six features are two too many to *look at* directly. **PCA** finds the orthogonal directions along which the cohort varies most, lets you keep the few that carry the signal, and hands you a 2-D picture you can actually read. Then comes the payoff: you will drop the **k-means clusters from 2.3** into that 2-D space and see whether the structure the algorithm found is *visible*, “reduce, then look.”

Ground rules. Implement PCA from scratch in NumPy/SciPy, centering, covariance, eigendecomposition (or SVD), projection. No scikit-learn, no PCA. The `info521` library gives you data, plotting, and self-checks only.

What’s a feature here. The same **six non-BP columns** `ds[“X”]`, standardized as in 2.3. PCA is unsupervised: the hypertension label is used only to *color* the projection, never to compute it.

What “Satisfactory” means for 2.4

- ☐ You implement **PCA from scratch** (center, covariance, **eigendecomposition or SVD**, project) and explain **variance explained**.
- ☐ You produce a **2-D projection / biplot** and read **which features load on PC1 and PC2**.
- ☐ You **integrate with 2.3**: re-view your **k-means clusters in PC space** and discuss what the projection reveals.
- ☐ **Play artifact** (below) complete and interpreted.

Setup

```
import numpy as np
from info521 import data, plotting, checks
```

```

plotting.use_course_style()
ds = data.load_clinical()

X_raw = ds["X"] # (n, 6): age, bmi, waist, chol, hdl, hba1c
feature_names = ds["features"]
labels, rule = data.hypertension(ds) # for COLORING the projection only

print(f"{X_raw.shape[0]} patients · {X_raw.shape[1]} features = {feature_names}")

```

0 · Center (and scale) the features

PCA is defined on **centered** data; because the six features are on different scales, standardize them first (so this is PCA on the correlation structure, which keeps `chol` from dominating purely by its units). Reuse your 2.3 `standardize`.

```
# TODO: Xz = standardized six-feature matrix (mean 0, sd 1 per column).
```

1 · PCA from scratch

Two equivalent routes, pick one and say why:

- **Covariance + eigendecomposition.** Form $\Sigma = \frac{1}{N-1} \mathbf{X}_c^\top \mathbf{X}_c$, eigendecompose $\Sigma = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$, sort by descending eigenvalue. Columns of $\mathbf{V}$ are the principal directions; λ_i is the variance along direction i .
- **SVD.** $\mathbf{X}_c = \mathbf{U} \mathbf{S} \mathbf{V}^\top$; the same $\mathbf{V}$, with $\lambda_i = s_i^2 / (N - 1)$. (Numerically steadier, note that in your writeup.)

Project with $\mathbf{Z} = \mathbf{X}_c \mathbf{V}_k$ for the top k components.

```

def pca_fit(Xc):
    # TODO: return (components V, explained_variance lambda) sorted descending.
    # Use np.linalg.eigh on the covariance OR np.linalg.svd on Xc. Be careful with
    # eigenvector ordering and sign (sign is arbitrary: fix a convention).
    raise NotImplementedError

def pca_project(Xc, V, k):
    # TODO: return the (n, k) projection onto the first k components.
    raise NotImplementedError

```

2 · Variance explained & the scree plot

The proportion of variance on component i is $\lambda_i / \sum_j \lambda_j$. Report the per-component and **cumulative** shares, and draw the **scree plot**.

```

def explained_variance_ratio(lambdas):
    # TODO: return lambdas normalized to sum to 1.
    raise NotImplementedError

# TODO: scree plot (per-component) + cumulative curve. How many components to reach,
# say, 80-90% of the variance? Is the six-feature cohort "really" low-dimensional?

```

3 · The biplot: what loads on PC1/PC2

Overlay the **feature loadings** (rows of **V** for PC1/PC2) on the 2-D score scatter. A feature's loading vector shows how strongly, and in which direction, it drives each axis.

```
# TODO: scatter the PC1-PC2 scores; draw each feature's loading as an arrow from the  
# origin. Which features load together on PC1? Does that echo the metabolic  
# (bmi/waist) and lipid (chol/hdl) blocks from the Part 1 data description?
```

4 · k-means clusters in PC space

Bring in your **best k-means labelling from 2.3** and color the PC1–PC2 scatter by cluster. This is the integration step: structure that lived in six invisible dimensions, now (partly) visible in two.

```
# TODO: color the 2-D projection by your 2.3 cluster labels; in a second panel,  
# color the SAME points by the hypertension label. Do the clusters occupy distinct  
# regions of PC space? Does either axis separate hypertensive from non-hypertensive?
```

Write 200–300 words: what does the projection reveal that the six-dimensional view hid? Connect three things, the features loading on PC1/PC2, the k-means clusters, and the hypertension label. If the clusters look clean in PC space but the label does not separate, what does that tell you about geometric vs. clinical structure (tying back to the 2.3 reveal)?

Play artifact: scree, biplot, clusters-in-PC-space

1. **Scree / cumulative variance:** how the intrinsic dimensionality reads off the eigenvalue spectrum.
2. **Biplot exploration:** rotate attention across components, what loads on PC3 that PC1/PC2 miss?
3. **Clusters-in-PC-space overlay:** the 2.3 clusters and the hypertension label, side by side, in the projection.

Write 150–250 words synthesizing the three panels into one claim about the cohort's structure.

```
# TODO: the three panels + the synthesizing interpretation.
```

i Gate B

2.3 and 2.4 together form the **unsupervised gate**. Per GRADING.md, Gate B is complete when **both** are Satisfactory and **Checkpoint 2.3** (the k-means / PCA checkpoint) is passed. There is no separate checkpoint for 2.4.

Looking ahead

In **2.5** you tie the whole semester together on this one cohort: regression (P1) → supervised hypertension classification (Gate A) → unsupervised structure (Gate B), and argue which framing answers which clinical question.